

LAB-5

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13.

AIM:

Create a Node.js application to view EMI Calculator

PROGRAM CODE:

Index.html:

```
<!DOCTYPE html>

<html>

<head>

<title>EMI Calculator</title>

</head>

<body>

<h2>EMI Calculator</h2>

<form action="/calculate" method="POST">

Loan Amount:

<input type="text" name="principal"><br><br>

Interest Rate (%):

<input type="text" name="rate"><br><br>
```

Loan Tenure (Years):

```
<input type="text" name="time"><br><br>
<button type="submit">Calculate EMI</button>
</form>
</body>
</html>
```

Server.js:

```
const express = require('express');
const bodyParser = require('body-parser');
const app = express();
app.use(bodyParser.urlencoded({ extended: true }));
app.use(express.static('public'));
app.post('/calculate', (req, res) => {
  let P = parseFloat(req.body.principal);
  let R = parseFloat(req.body.rate) / 12 / 100;
  let N = parseFloat(req.body.time) * 12;
  let EMI = (P * R * Math.pow(1 + R, N)) /
    (Math.pow(1 + R, N) - 1);
  res.send("Monthly EMI = ₹ " + EMI.toFixed(2));
});
app.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

EXECUTION STEPS:

Step 1: Install Node.js

Check installation:

`node -v`

`npm -v`

Step 2: Create Project Folder

`mkdir emi-calculator`

`cd emi-calculator`

Step 3: Initialize Node Project

`npm init -y`

Step 4: Install Express

`npm install express body-parser`

Step 5: Project Structure

emi-calculator

|

├── server.js

├── package.json

└── public

 └── index.html

Step 6: Create server.js

Step 7: Create public/index.html

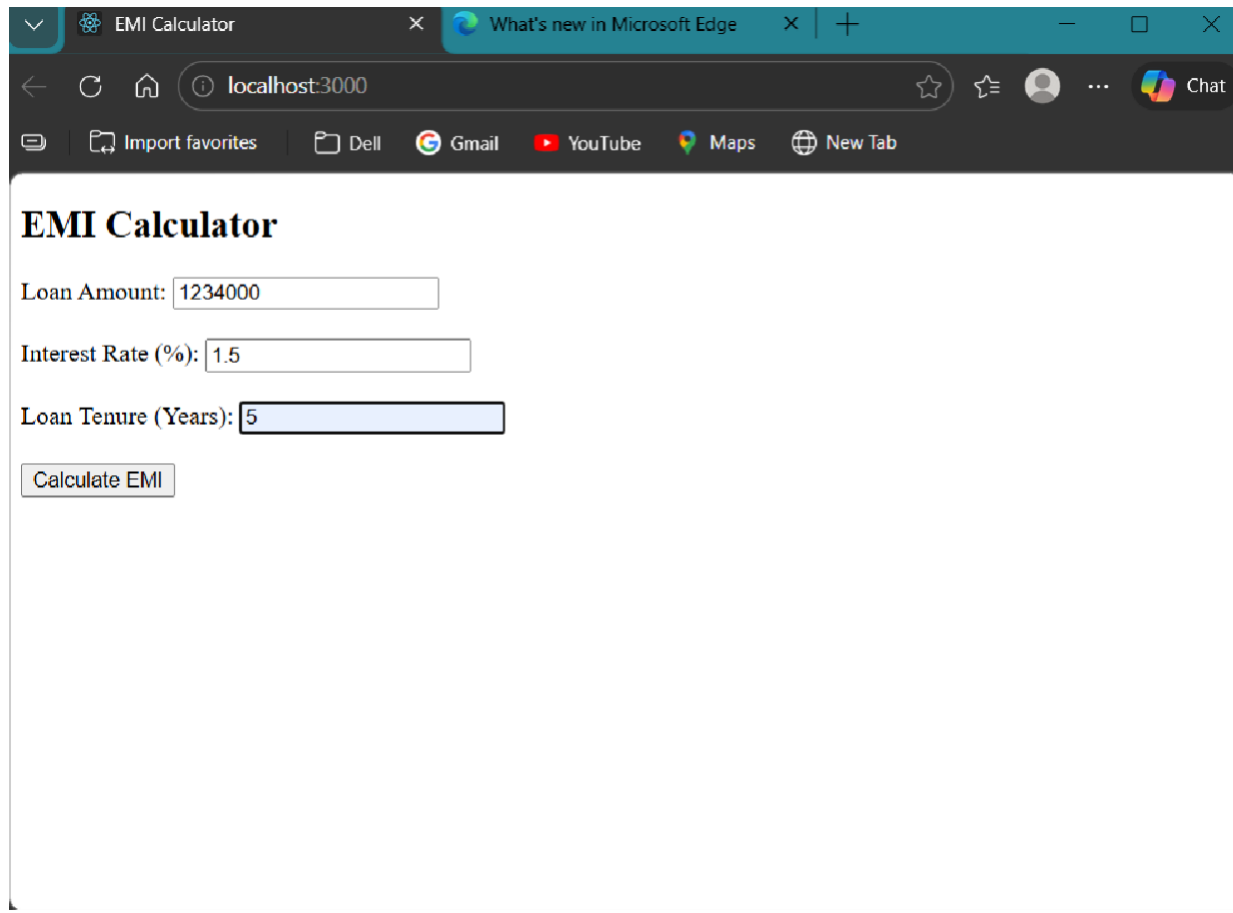
Step 8: Run Program

`node server.js`

Open browser:

`http://localhost:3000`

OUTPUT SCREENSHOT:



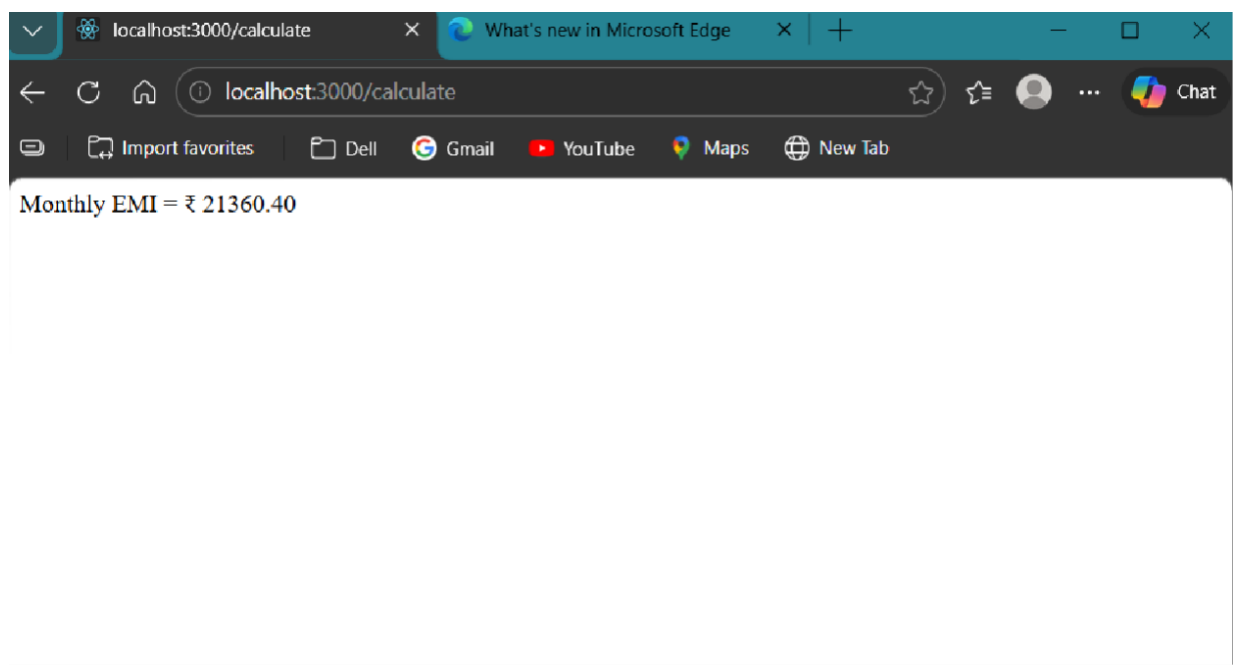
The screenshot shows a web browser window with the title "EMI Calculator". The address bar displays "localhost:3000". The page contains three input fields: "Loan Amount" with the value "1234000", "Interest Rate (%)" with the value "1.5", and "Loan Tenure (Years)" with the value "5". Below these fields is a button labeled "Calculate EMI".

EMI Calculator

Loan Amount:

Interest Rate (%):

Loan Tenure (Years):



The screenshot shows the same web browser window, but the address bar now displays "localhost:3000/calculate". The page content has changed to show the result: "Monthly EMI = ₹ 21360.40".

Monthly EMI = ₹ 21360.40

14.

AIM:

Create a TODO application in Node.js and deploy it to GitHub.

PROGRAM CODE:

Index.html:

```
<!DOCTYPE html>

<html>

<head>

<title>TODO App</title>

</head>

<body>

<h2>TODO Application</h2>

<form action="/add" method="POST">

Enter Task:

<input type="text" name="task">

<button type="submit">Add Task</button>

</form>

</body>

</html>
```

Server.js:

```
const express = require('express');

const bodyParser = require('body-parser');

const app = express();
```

```
let todos = [];  
app.use(bodyParser.urlencoded({ extended: true }));  
app.use(express.static('public'));  
app.post('/add', (req, res) => {  
  let task = req.body.task;  
  todos.push(task);  
  let list = "<h2>TODO LIST</h2>";  
  todos.forEach((t,i)=>{  
    list += (i+1) + ". " + t + "<br>";  
  });  
  res.send(list + "<br><a href='/'>Back</a>");  
});  
app.listen(3000, () => {  
  console.log("Server running at http://localhost:3000");  
});
```

EXECUTION STEPS:

Step 1: Install Node.js

Check installation:

node -v

npm -v

Step 2: Create Project Folder

mkdir todo-app

cd todo-app

Step 3: Initialize Node Project

npm init -y

Step 4: Install Express

```
npm install express body-parser
```

Step 5: Project Structure

```
todo-app
|
├── server.js
└── public
    └── index.html
```

Step 6: Create server.js

Step 7: Create public/index.html

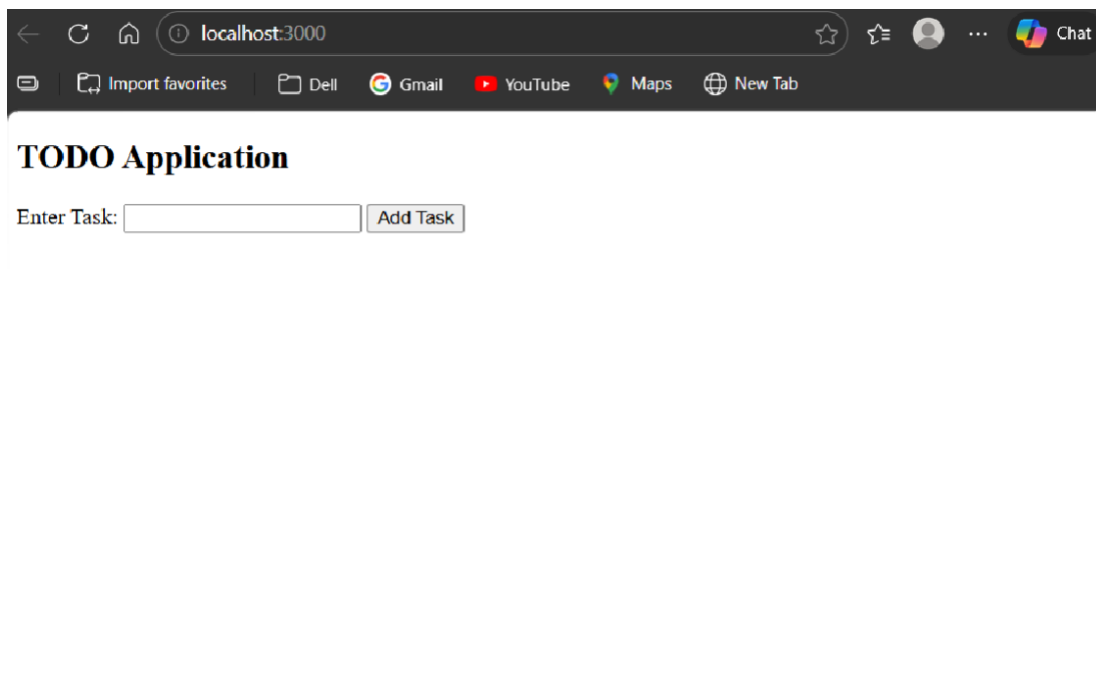
Step 8: Run Program

```
node server.js
```

Open browser:

<http://localhost:3000>

OUTPUT SCREENSHOT:



TODO LIST

1. complete assignment

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AIM:

Design a Hotel Bill Screen using Node.js

PROGRAM CODE:

Index.html:

```
<!DOCTYPE html>

<html>

<head>

<title>Hotel Billing</title>

</head>

<body>

<h2>Hotel Billing System</h2>

<form action="/bill" method="POST">

Select Item:

<select name="item">
```



```
<option>Dosa</option>
<option>Idli</option>
<option>Biryani</option>
<option>Tea</option>
</select>
<br><br>
Quantity:
<input type="number" name="quantity">
<br><br>
<button type="submit">Generate Bill</button>
</form>
</body>
</html>
```

Server.js:

```
const express = require('express');
const bodyParser = require('body-parser');
const app = express();
app.use(bodyParser.urlencoded({ extended: true }));
app.use(express.static('public'));
app.post('/bill', (req, res) => {
  let item = req.body.item;
  let quantity = parseInt(req.body.quantity);
  let price = 0;
  if(item=="Dosa") price=40;
  if(item=="Idli") price=30;
```

```

    if(item=="Biryani") price=120;
    if(item=="Tea") price=15;
    let total = price * quantity;
    res.send(`
        <h2>Hotel Bill</h2>
        Item : ${item}<br>
        Quantity : ${quantity}<br>
        Price : ${price}<br>
        Total Amount : ₹${total}
        <br><br>
        <a href="/">Back</a>
    `);
});

app.listen(3000,()=>{
    console.log("Server running on http://localhost:3000");
});

```

EXECUTION STEPS:

Step 1: Install Node.js

Check installation:

node -v

npm -v

Step 2: Create Project Folder

mkdir hotel-bill

cd hotel-bill

Step 3: Initialize Node Project

```
npm init -y
```

Step 4: Install Express

```
npm install express body-parser
```

Step 5: Project Structure

```
hotel-bill
|
├── server.js
└── public
    └── index.html
```

Step 6: Create server.js

Step 7: Create public/index.html

Step 8: Run Program

```
node server.js
```

Open browser:

<http://localhost:3000>

OUTPUT SCREENSHOT:

Hotel Billing System

Select Item:

Quantity:

Hotel Bill

Item : Dosa
Quantity : 10
Price : 40
Total Amount : ₹400

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